

MURMANSK, Russian Federation

WMO: 221130

Lat: **68.9558N**Lon: **33.0428E**Elev: **266**StdP: **14.56**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-21.2	-34.2	0.8	-27.1	-27.2	1.2	-20.8	28.4	19.5	24.6	21.3	3.2	180	0.766

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	76.7	61.4	71.8	59.0	67.9	57.1	63.2	72.8	60.6	69.2	58.4	66.4	8.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.2	75.7	67.0	56.8	69.3	64.6	54.0	62.7	61.7	28.7	73.6	27.0	69.5	25.4	66.4	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.4	19.0	16.4	DB	-31.6	82.4	7.6	4.8	-37.1	85.9	-41.6	88.6	-45.9	91.3	-51.4	94.8	(4)
(5)			WB	-31.9	66.7	7.6	3.1	-37.3	68.9	-41.7	70.8	-46.0	72.5	-51.5	74.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	32.1	10.3	10.4	19.3	29.9	40.1	49.2	56.1	52.3	44.8	33.9	21.7	15.8
	DBStd	18.70	14.50	14.47	10.97	8.34	7.60	7.62	7.31	6.41	6.16	7.91	11.49	13.01
	HDD50	6979	1230	1109	951	604	322	106	23	46	179	501	849	1059
	HDD65	12026	1695	1529	1416	1054	772	477	291	398	606	965	1299	1524
	CDD50	448	0	0	0	0	15	82	211	117	23	0	0	0
	CDD65	21	0	0	0	0	0	3	14	4	0	0	0	0
	CDH74	238	0	0	0	0	6	35	161	36	0	0	0	0
	CDH80	39	0	0	0	0	1	5	30	3	0	0	0	0
Wind	WSAvg	7.1	6.6	7.1	7.6	7.6	7.9	7.8	7.0	6.2	6.9	7.0	6.6	6.9
Precipitation	PrecAvg	19.1	1.2	0.9	0.9	0.9	1.3	2.1	2.5	2.4	2.0	1.9	1.5	1.4
	PrecMax	25.9	2.2	2.0	2.1	2.8	2.7	5.6	5.8	7.0	4.7	4.6	4.1	4.4
	PrecMin	10.9	0.4	0.1	0.2	0.2	0.4	0.1	0.7	0.8	0.7	0.4	0.4	0.1
	PrecStd	3.2	0.5	0.4	0.5	0.4	0.6	1.1	1.2	1.1	0.9	0.9	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	38.7	43.0	53.9	70.1	78.8	84.1	78.7	66.2	53.5	43.0	39.3
		MCWB	34.8	35.4	36.0	43.5	53.3	60.6	64.7	63.2	56.2	48.3	40.8	36.2
	2%	DB	33.7	35.2	38.9	47.7	62.8	71.8	78.6	72.9	61.9	49.8	39.0	36.8
		MCWB	32.0	32.6	34.5	40.0	49.6	57.0	62.0	60.3	53.8	46.0	37.3	34.5
	5%	DB	30.9	32.4	35.7	44.3	58.3	67.9	73.8	67.8	58.6	46.7	36.8	34.2
		MCWB	29.4	30.3	32.0	38.0	47.7	54.8	60.4	58.1	52.6	43.8	35.0	32.4
	10%	DB	28.4	29.9	33.5	41.1	53.3	63.0	69.8	63.8	55.1	44.5	35.1	31.9
		MCWB	27.0	28.0	30.5	36.0	45.0	52.6	59.1	56.6	50.5	42.1	33.6	30.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	35.6	36.9	44.3	55.2	63.1	67.4	65.1	57.4	49.2	40.8	36.5
		MCDB	36.5	37.8	41.5	53.1	69.0	74.7	77.8	75.2	63.8	51.9	42.1	38.6
	2%	WB	32.0	32.7	34.6	40.5	51.1	58.9	64.4	61.6	54.9	46.3	37.2	34.5
		MCDB	33.2	34.7	38.1	45.9	61.5	69.4	74.5	69.8	60.3	49.1	38.5	36.3
	5%	WB	29.4	30.5	32.4	38.4	47.9	56.1	62.0	59.3	52.8	43.9	34.9	32.6
		MCDB	30.7	32.6	35.1	43.1	56.9	65.6	71.2	66.5	57.6	46.1	36.1	34.1
	10%	WB	26.9	27.9	30.6	36.4	45.0	53.5	59.7	57.1	50.7	41.8	33.3	30.2
		MCDB	28.2	29.6	33.1	40.3	52.3	62.6	68.3	63.0	54.7	44.1	34.5	31.6
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.0	17.5	15.1	13.5	15.0	15.9	14.8	12.8	9.1	12.4	15.2
		MCDBR	12.5	11.6	13.1	17.3	24.0	23.0	24.6	22.5	18.8	10.0	7.2	10.5
		MCWBR	11.9	10.4	10.4	11.4	13.5	11.7	12.2	12.1	12.3	7.9	6.7	9.8
	5% WB	MCDBR	12.8	11.4	11.6	15.5	22.3	21.2	22.5	19.4	16.6	9.1	6.9	10.4
		MCWBR	12.5	10.4	9.7	10.6	13.0	12.0	12.6	11.4	11.3	7.7	6.8	9.9
Clear-Sky Solar Irradiance	taub	0.206	0.254	0.301	0.332	0.354	0.350	0.368	0.349	0.318	0.277	0.361	N/A	
	taud	1.911	2.046	2.098	2.154	2.339	2.446	2.420	2.449	2.497	2.226	2.673	N/A	
	Ebn at Noon	12	156	222	253	264	269	259	250	228	149	24	0	
	Edh at Noon	2	14	26	34	32	30	30	26	19	12	3	0	
All-Sky Solar Radiation	RadAvg	6	116	520	1042	1362	1488	1385	945	520	151	20	0	
	RadStd	0	14	30	76	154	121	164	100	55	18	2	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page